

Resolving Wigner Negativity, Semiclassical Caustics, and Schrödinger Nodes via the Uncertainty Principle

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Wigner negativity, semiclassical caustics, and Schrödinger nodes follow from ignoring Heisenberg constraints — equivalently, from assuming unbounded classical action. Enforcing the action capacity of a physical system via symplectic phase-space cells resolves all three simultaneously. The resulting phase-space densities are provably non-negative, free of turning-point divergences, and strictly positive throughout the classically allowed region.

The Wigner–Weyl distribution [1] is an exact phase-space representation of a quantum state, and it takes negative values for any state beyond the ground state. The WKB probability density [2, 3] is the leading-order semiclassical approximation, and it diverges at every classical turning point. The exact wavefunction $|\psi_n(q)|^2$ is exactly zero at every wavefunction node. Each is treated as a feature of its own formalism. They share a kinematic root: each requires evaluating a density at a point in phase space, which by the uncertainty principle requires $\delta q \rightarrow 0$, hence $\Delta p \rightarrow \infty$, hence unbounded classical action. No physical system carries unbounded action.

The conventional remedies treat the three separately. The Husimi Q -function [4] convolves W with a fixed coherent-state kernel of unit width, recovering pointwise non-negativity at the cost of erasing every scale below the ground-state cell. Langer’s connection formulas [2, 3] patch the WKB wavefunction across each turning point in an asymptotic expansion local in q ; the result is a wavefunction in three pieces, and Wigner-transforming it returns the negativity. Node-counting arguments [?] treat the exact zeros as a structural feature of the spectrum, not as an artifact requiring resolution. None of the three remedies addresses the others.

Here we replace the Husimi kernel with one whose width is fixed by the state’s own action capacity. The construction is forced by three results from symplectic geometry: de Gosson’s mapping of the Heisenberg limit onto a Fermi blob [6, 7], Zurek’s reciprocal scales [5], and Gromov’s non-squeezing theorem [8]. The kernel carries phase-space area $h/2$, saturating the Heisenberg bound by construction. Hudson’s theorem [9] guarantees the resulting density is non-negative everywhere. The kernel’s finite momentum support guarantees it is finite everywhere. The same kinematic constraint that excludes pointwise infinity excludes pointwise zero, and the resolved density is strictly positive in the classically allowed region. One construction handles all three pathologies.

The action capacity of a quantum state is read directly from its Robertson–Schrödinger covariance [?]. The second moments $\sigma_{qq}, \sigma_{pp}, \sigma_{qp}$ define classical semi-axes $\Delta q = \sqrt{2\sigma_{qq}}, \Delta p = \sqrt{2\sigma_{pp}}$, projecting the 1σ contour

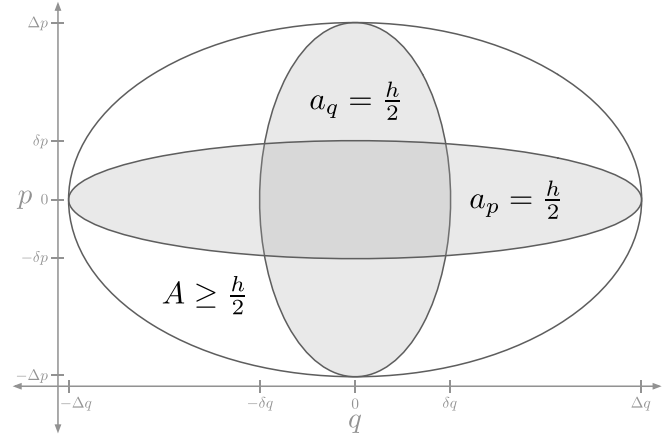


FIG. 1. **The symplectic quantum of action** (schematic, 1D case). The Heisenberg limit $\sigma_q \sigma_p \geq \hbar/2$ admits classical semi-axes $\Delta q = \sqrt{2\sigma_q}, \Delta p = \sqrt{2\sigma_p}$; de Gosson’s construction projects the 1σ contour onto a Fermi blob of phase-space area $A = \pi \Delta q \Delta p \geq h/2$. Its interior contains a conjugate pair of de Gosson’s quantum blobs, $a_q = \pi \delta q \Delta p = h/2$ and $a_p = \pi \Delta q \delta p = h/2$, summing to the quantum of action $h = a_q + a_p$. Their minor semi-axes are Zurek’s reciprocal scales $\delta q = \hbar/\Delta p$ and $\delta p = \hbar/\Delta q$; a_q is squeezed along the position axis and a_p along the momentum axis, and their joint structure is accessible only through ensemble tomography on separate copies of the state. As A grows, a_q and a_p compress along their conjugate axes, sharpening phase-space resolution.

onto a Fermi blob [7] of phase-space area

$$A = \pi \Delta q \Delta p \geq \frac{h}{2}. \quad (1)$$

The Heisenberg limit imposes $A \geq h/2$, with the lower bound saturated by minimum-uncertainty Gaussians. Zurek’s reciprocal scales [5] partition the blob’s interior: $\delta q = \hbar/\Delta p, \delta p = \hbar/\Delta q$. These define a conjugate pair of orthogonal quantum blobs [6],

$$a_q = \pi \delta q \Delta p = \frac{h}{2}, \quad a_p = \pi \Delta q \delta p = \frac{h}{2}, \quad (2)$$

each carrying the irreducible area $h/2$, a_q squeezed along the position axis and a_p along the momentum axis (Fig. 1). Holding their areas fixed while Δq and Δp expand forces the reciprocal compression of δq and δp [8];

conjugate pairing locks their sum to Planck's quantum, $a_q + a_p = h$. The squeezing structure is operational: states with δq or δp below the ground-state scale are the basis of quantum metrology beyond the standard quantum limit [? ?].

The action capacity A and the orbit's Bohr–Sommerfeld action $A_{\text{BS}} = \oint p dq$ are different objects. Both are areas in classical phase space with units of action; they measure different geometric content. For the harmonic oscillator the two coincide and the Fermi blob equals the energy-shell ellipse. For anharmonic potentials they separate, with $A \leq A_{\text{BS}}$ in general. The kernel widths δq and δp are governed by A , not by A_{BS} : the state's actual phase-space extent, not the orbit it would trace if it had no quantum spread, sets the resolution scales.

The Gaussian kernels of phase-space area $h/2$ corresponding to the conjugate quantum blobs are

$$G_{\delta q}(q, p) = \frac{1}{\pi} \exp\left(-\frac{q^2}{\delta q^2} - \frac{p^2}{\Delta p^2}\right), \quad (3)$$

$$G_{\delta p}(q, p) = \frac{1}{\pi} \exp\left(-\frac{q^2}{\Delta q^2} - \frac{p^2}{\delta p^2}\right). \quad (4)$$

$G_{\delta q}$ is squeezed along the position axis (width δq) and spans Δp in momentum; $G_{\delta p}$ is the conjugate. Each saturates the symplectic-capacity bound at $h/2$. Convolution produces a phase-space density,

$$P_{\delta q}(q, p) = (W * G_{\delta q})(q, p), \quad (5)$$

that is non-negative everywhere by Hudson's theorem [9], since each kernel attains the symplectic positivity condition $\lambda_{\min}(\Sigma + \frac{i}{2}J) = 0$. The kernel's finite momentum support of width Δp simultaneously bounds the kinematic action at the classical turning points, replacing the WKB caustic with a finite-amplitude peak. Marginalizing the convolved 2D density over p ,

$$\rho_{\delta q}(q) = \int (W_{\text{cl}} * G_{\delta q})(q, p) dp, \quad (6)$$

where $W_{\text{cl}}(q, p) \propto \delta(H - E)$ is the classical microcanonical density, gives a 1D position density that is everywhere finite even where P_{WKB} diverges and strictly positive in the classically allowed region.

The pipeline that produces $\rho_{\delta q}$ requires no wavefunction. From the Bohr–Sommerfeld energy of an eigenstate, the classical orbit's turning points fix the moments Δq and Δp , which fix δq and δp via the reciprocal scales, which fix the kernel. The kernel is orbit-derived, not state-derived: the same $G_{\delta q}$ that resolves the classical microcanonical density into $\rho_{\delta q}$ resolves the quantum Wigner function into $P_{\delta q}$. One construction handles both inputs.

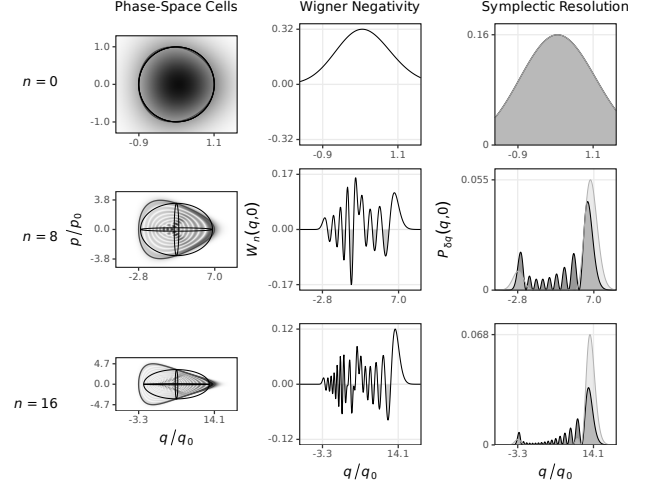


FIG. 2. **Morse oscillator, Wigner.** Rows show eigenstates $n = 0, 8, 16$. *Left:* Phase-space Wigner distribution $W_n(q, p)$ with the symplectic quantum of action overlaid. The Fermi blob becomes increasingly anisotropic as n grows, reflecting the asymmetric Morse confinement. *Center:* Cross-section $W_n(q, 0)$ showing the growing oscillation structure; negative regions deepen with n . *Right:* Cross-section $P_{\delta q}(q, 0)$, non-negative everywhere by Hudson's theorem, with the Husimi cross-section $Q(q, 0)$ overlaid as comparator. The symplectic kernel preserves the multi-lobe envelope of $W_n(q, 0)$; the fixed-width Husimi kernel collapses the interior structure to a single peak biased toward the slow turning point.

For the Morse oscillator $V(q) = D_e(1 - e^{-\alpha q})^2$ with $D_e = 12.5$, $\alpha = 0.2$, $m = 1$, the natural frequency $\omega_0 = \alpha\sqrt{2D_e/m} = 1$ matches the harmonic oscillator's. Bound states have $A < A_{\text{BS}}$, with the gap widening as the orbit reaches further into the soft-wall region. Figure ?? shows the Wigner-input pipeline at $n = 0, 8, 16$. The exact cross-section $W_n(q, 0)$ develops $n + 1$ alternating-sign lobes whose envelope tracks the asymmetric orbit. The symplectic-resolved cross-section is non-negative by Hudson's theorem and retains the full multi-lobe structure. The Husimi cross-section, overlaid for comparison, smooths the structure into a single peak biased toward the soft-wall side; its kernel width $q_0 = 1$ exceeds the orbit's natural δq for $n \geq 8$, and the contrast grows with n .

Figure ?? shows the semiclassical-input pipeline for the same eigenstates. The WKB density $|\psi_{\text{WKB}}|^2$ in the center column carries the canonical $n + 1$ -lobed Bohr–Sommerfeld interference pattern — the half-orbit phase advances by $\pi(n + 1/2)$, and $\cos^2$ of that phase has n zeros and $n + 1$ peaks — together with the textbook caustic divergences at the turning points where $p \rightarrow 0$. The right column shows $\rho_{\delta q}$ with the Langer/Miller uniform Airy density as the prior-art comparator. The two agree on the location, count, and envelope of the oscillation lobes. The Airy density is more peaked because

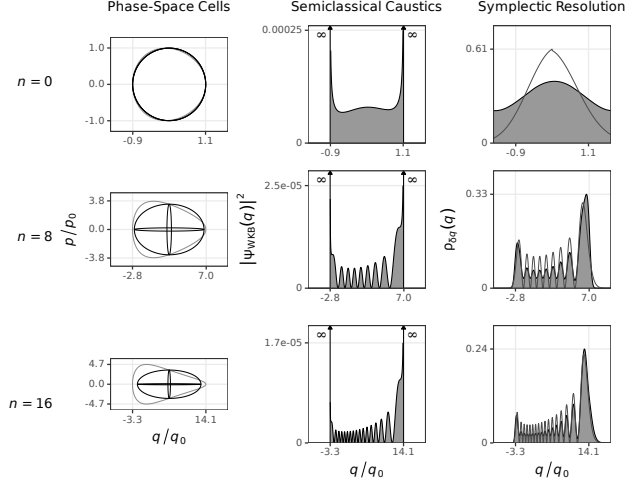


FIG. 3. **Morse oscillator, semiclassical.** Same eigenstates as Fig. ?? . *Left:* Classical orbit at energy E_n with the symplectic quantum of action overlaid. The orbit deforms from a circle ($n = 0$) to a horseshoe ($n = 16$) as it explores the soft-wall region. *Center:* Oscillating WKB density $|\psi_{\text{WKB}}(q)|^2 = (A^2/p) \cos^2(S/\hbar - \pi/4)$, with $n + 1$ interference lobes between the turning points and divergences (rendered as infinity arrows) at the turning points where $p \rightarrow 0$. *Right:* Symplectic resolution $\rho_{\delta q}(q)$ with the Langer/Miller uniform Airy density [3] overlaid as the prior-art comparator. The $n + 1$ oscillation lobes are preserved; the turning-point divergences are replaced by finite peaks. The Airy curve has higher peaks and deeper troughs than $\rho_{\delta q}$ at every n because the Langer construction is evaluated at infinite resolution; the symplectic resolution is the same content coarse-grained at the orbit's natural scale $\delta q = \hbar/\Delta p$.

the Langer construction is evaluated at infinite resolution: peaks taller than the symplectic curve and troughs that touch zero. The symplectic resolution is the same content coarse-grained at the orbit's natural scale δq ; below δq both peaks and troughs are coarse-grained away, and the envelope is what remains.

The Morse ground state ($n = 0$) illustrates a structural property of the construction. Reading the momentum-product directly from the Bohr-Sommerfeld classical orbit yields $\Delta q \Delta p < \hbar/2$ at low quantum numbers in anharmonic potentials [?]. The orbit can do this because it is a classical object, not a quantum state; the orbit alone is not subject to Heisenberg. The action capacity, however, is fixed-resolution by construction: the smallest cell it can render has area $\hbar/2$, set by Gromov's non-squeezing theorem [8] and de Gosson's quantum-blob structure [6]. When the orbit reads sub-resolution, the QoA returns the floor blob, with Δq and Δp scaled uniformly to saturate $A = \hbar/2$ while preserving the orbit's aspect ratio. No correction or override is involved: the QoA reports what the QoA can resolve, in the same way an optical microscope's resolution is set by its aperture rather than by the size of any sub-resolution feature ly-

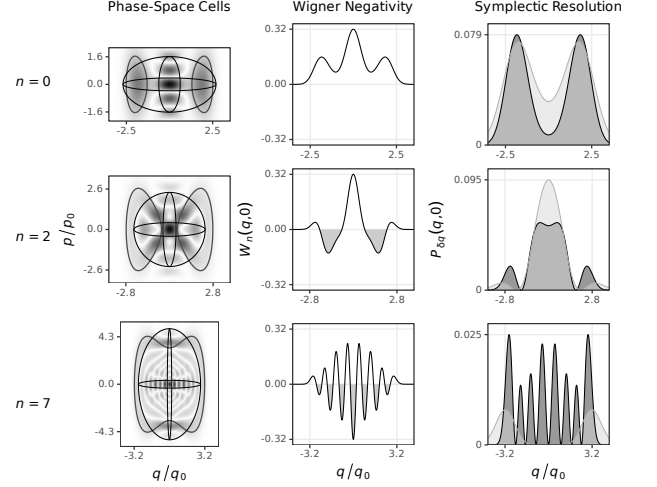


FIG. 4. **Symmetric double well, Wigner.** Rows show eigenstates $n = 0$ (lower symmetric doublet partner), $n = 2$ (upper symmetric doublet partner, still below the barrier), and $n = 7$ (above-barrier connected orbit). *Left:* Phase-space Wigner distribution showing the inter-well interference fringes characteristic of tunneling doublets and the peanut-shaped phase-space orbit of an above-barrier state. *Center:* Cross-section $W_n(q, 0)$. The central peak in the doublet rows is the phase-space signature of coherent superposition between the two wells. *Right:* Cross-section $P_{\delta q}(q, 0)$ with Husimi $Q(q, 0)$ overlaid. The symplectic kernel preserves the three-peak structure (left well, central inter-well coherence, right well); the Husimi kernel flattens the central peak.

ing under the lens. Such a floor is a necessary feature of any phase-space construction that cannot dip below the Heisenberg bound.

For the symmetric double well $V(q) = -\mu^2 q^2/2 + \lambda q^4/4$ with $\mu^2 = 4$, $\lambda = 1$, the barrier height is $V_0 = \mu^4/(4\lambda) = 4$ and the well minima sit at $q = \pm 2$. Sub-barrier states form symmetric/antisymmetric tunneling doublets; above-barrier states have a single connected orbit through the barrier region. Figure ?? shows the Wigner-input pipeline at the lower doublet partner ($n = 0$), the upper doublet partner still below the barrier ($n = 2$), and a connected above-barrier state ($n = 7$). The cross-section $W_n(q, 0)$ for the doublet partners shows three peaks: the two well peaks plus a central interference peak that is the phase-space signature of coherent superposition between the wells. This central peak survives the symplectic resolution. It does not survive the Husimi resolution: the fixed coherent-state kernel of width $q_0 = 1$ exceeds the inter-peak separation and flattens the central peak into the wells. The double-well's lowest doublet is the most stringent test of any phase-space resolution method, because the central peak is small in amplitude and narrow in extent — and the symplectic kernel resolves it precisely because $\delta q = \hbar/\Delta p$ is small enough when Δp , set by the doublet's delocalized covariance, is

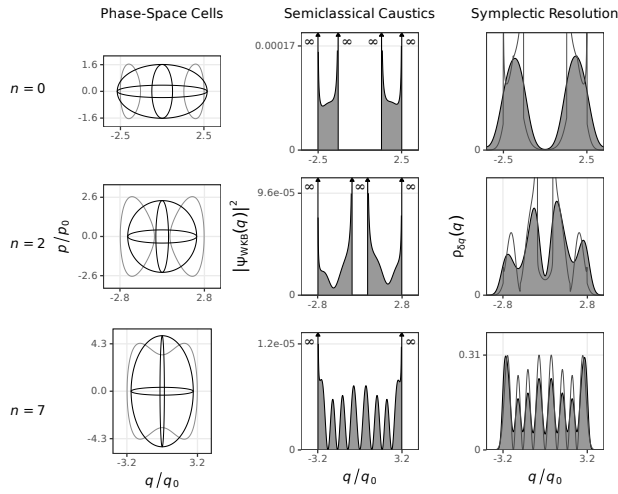


FIG. 5. **Symmetric double well, semiclassical.** Same eigenstates as Fig. ?? . *Left:* Classical orbit at energy E_n with the symplectic quantum of action overlaid. Sub-barrier states have two disconnected orbital loops; the above-barrier state has a single peanut. *Center:* Oscillating WKB density. Sub-barrier states have four turning points (and four caustic divergences); above-barrier states have two. *Right:* Symplectic resolution $\rho_{\delta q}(q)$ with the per-orbit Langer/Miller Airy density overlaid. The per-well Langer construction has known peaks at the inner turning points (Airy lobe maxima at the well's classical boundary) that clip the panel; the symplectic resolution is everywhere finite and resolves the two-well structure smoothly. For the above-barrier state the two methods agree on the multi-lobe interior.

large.

Figure ?? shows the semiclassical-input pipeline. The WKB density for sub-barrier states has four turning points and four caustic divergences, two per well; the symplectic-resolved $\rho_{\delta q}$ replaces them with finite peaks. The right column shows the per-orbit Langer/Miller Airy density: a textbook construction that builds the uniform Airy wavefunction on each well separately and sums them. For sub-barrier doublets the construction reports tall peaks at every inner turning point — standard Langer behavior at orbit boundaries, applied here to a state whose true wavefunction has its mass at the well minima rather than at the inner walls. The Airy curves clip out the top of the panel; the symplectic curve does not. For the above-barrier state the orbit is connected and Langer behaves normally; the two methods agree on lobe count and amplitude. Comparing rows: the symplectic kernel returns the same well-localized two-peak structure for sub-barrier classical input as it does for above-barrier input, with no inter-well peak — because the classical microcanonical density carries no inter-well phase information. The inter-well peak in the Wigner figure is genuinely quantum, and the kernel preserves it when given quantum input while not synthesizing it from classical input.

The same kinematic root that excludes pointwise infinity excludes pointwise zero. A density that takes the value ∞ at a point requires concentrating finite probability into zero width: $\delta q \rightarrow 0$, hence $\Delta p \rightarrow \infty$, hence unbounded action. A density that takes the value 0 at a point requires resolving the field to a single point: $\delta q \rightarrow 0$ again, hence the same unbounded action. The exact zeros of $|\psi_n(q)|^2$ at the wavefunction's nodes are the reciprocal of the WKB infinity at the turning points, and neither survives at any finite resolution. The action capacity sets a finite scale $\delta q = \hbar/\Delta p$ too coarse to support either; the resolved density is everywhere finite and strictly positive in the classically allowed region.

The construction has no free parameters. Δq and Δp are read directly from the orbit; δq and δp follow from the reciprocal scales at the Heisenberg limit; the kernel is fixed. Across the Morse oscillator and the symmetric double well the kernel preserves quantum oscillation structure on scales between δq and q_0 that the fixed-width Husimi kernel smooths, and removes the turning-point divergences and node zeros that the Langer/Airy and exact-Schrödinger constructions report at infinite resolution. The kernel-capacity ratio $q_0/\delta q$ sets the magnitude of the Husimi contrast; for high-capacity states this ratio is large, and the structure preserved is correspondingly rich. Wigner negativity, semiclassical caustics, and Schrödinger nodes are resolved by enforcing the action capacity that the uncertainty principle imposes on every physical state.

All numerical data, figure-generating code, and reproducibility scaffolding for this paper are available at <https://doi.org/10.17605/OSF.IO/EV8H6>.

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